CODING FROM COGNITIVE (NEURO)SCIENCE

& reprogramming stereoytpes!

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Do I *really need* to learn how to program as a researcher in psychology / cognitive science?

✓ **Definitively YES!**

For me:

- **Control, flexibility, automaticity.**

For others:

- **Precision, transparency, reproducibility**

Why is it *so hard* to access programing as a women in cognitive science?

When we think about programmers, developers, computer scientists…

- Male, probably white
- ***Brilliant***, great intelligence
- Socially awkward
- Narrow, focused interests
- Nerd / Geek

[Cheryan et al., 203; Pantic et al., 2018]

Please draw a highly skilled professional programmer.

When we think about **programmers, developers, computer scientists**…

- **Male, probably white**
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The **stereotypical, male-dominated view** of computer science **exceeds** other STEM domains and **emerges early** (6-8 years old) [e.g., Miller et al., 2024; Master et al., 2025]

When we think about psychologists or psychology…

Woman!
Extremely feminized field

Prosocial, empathetic, care-giver

Soft science (or not a science), overall *easy*

[Boysen et al., 2021; Brinthaupt et al., 2016; Alperin et al., 1985]

Please draw a highly skilled professional psychologist.

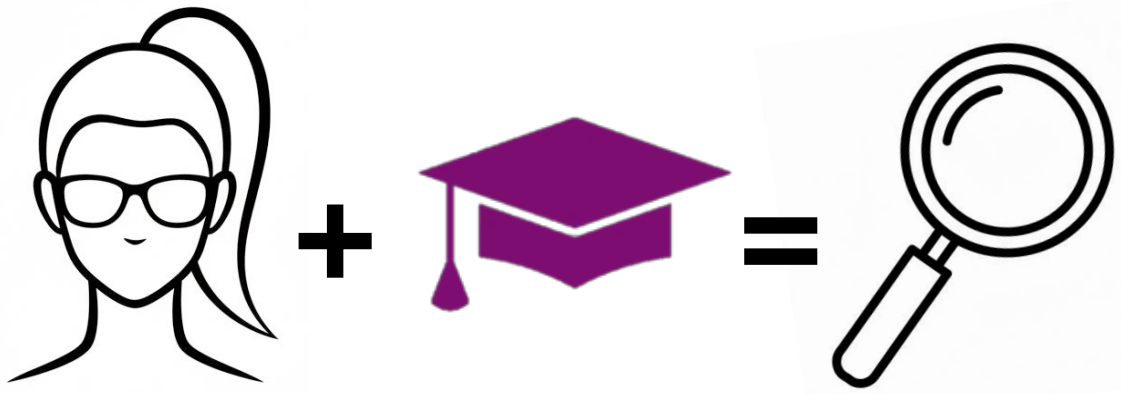
When we think about psychologists or psychology…

Woman!
Extremely feminized field

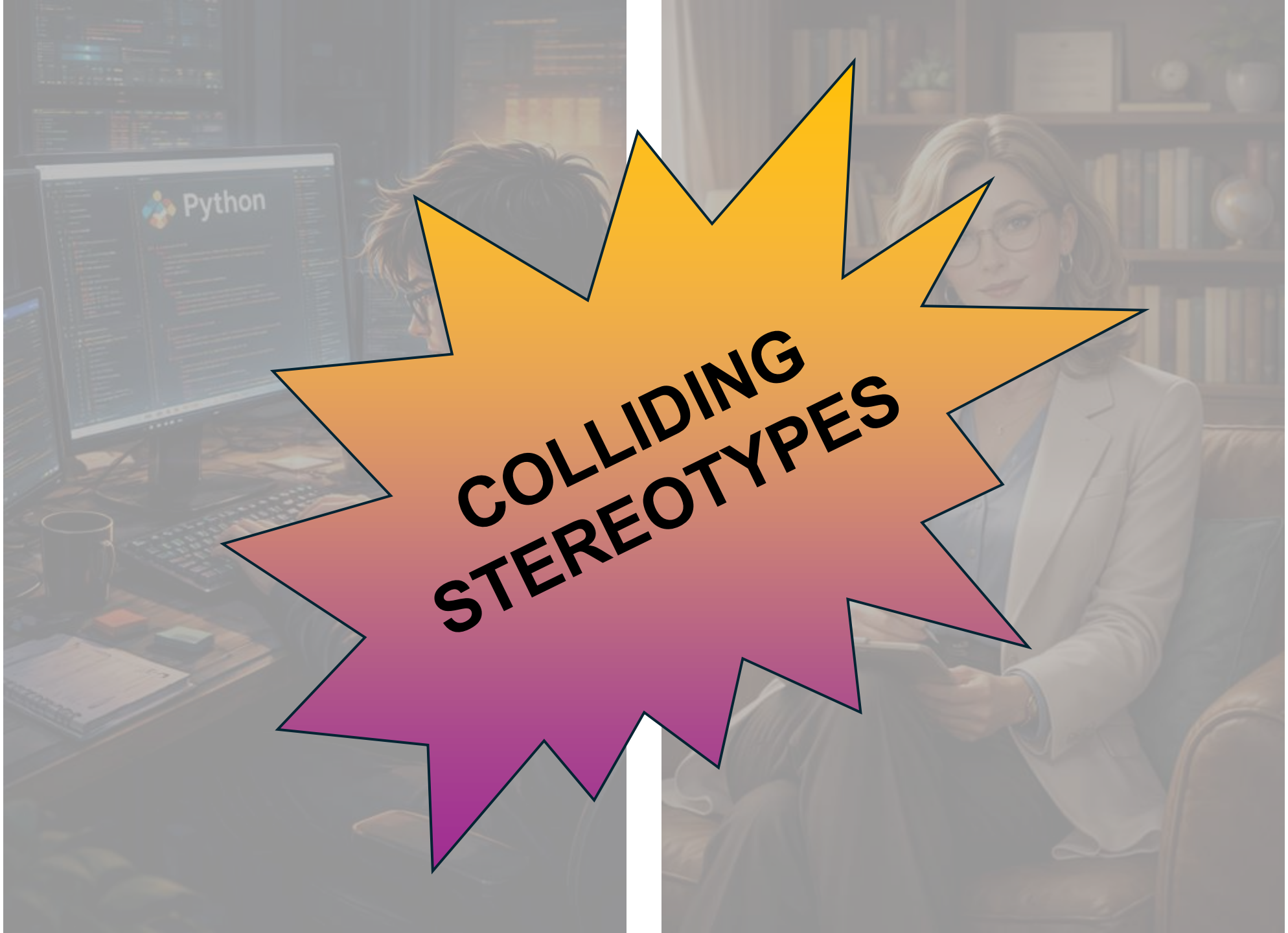
Prosocial, empathetic, care-giver

Soft science (or not a science), overall *easy*

[Boysen et al., 2021; Brinthaupt et al., 2016; Alperin et al., 1985]



Python


Python
COLLIDING
STEREOTYPES

TOP-DOWN (STRUCTURAL) EFFECTS

"Psychologist won't learn easily how to program", "Coding is useless for a psychologist"

At many universities, programming is not part of the Psychology curriculum.

Psychology students don't get expose to programming

Reduced likelihood of developing coding skills

Sense of belonging, expectations of success, ability self-concept, interest in the field…

"I've never been good with computers", "Programming seems super difficult", "It's not for me"

BOTTOM-UP (INDIVIDUAL) EFFECTS

Why is it *so hard* to access programing as a women in cognitive science?

- Pervasive gender gap in computational skills
- Polarized, colliding stereotypes!

Socially shared (and perpetuated) stereotypes shape decision-making processes both at the individual and institutional levels.

WHAT *can we do* to foster access to programming from psychology / cognitive science?

1. **Early exposure to programming.**

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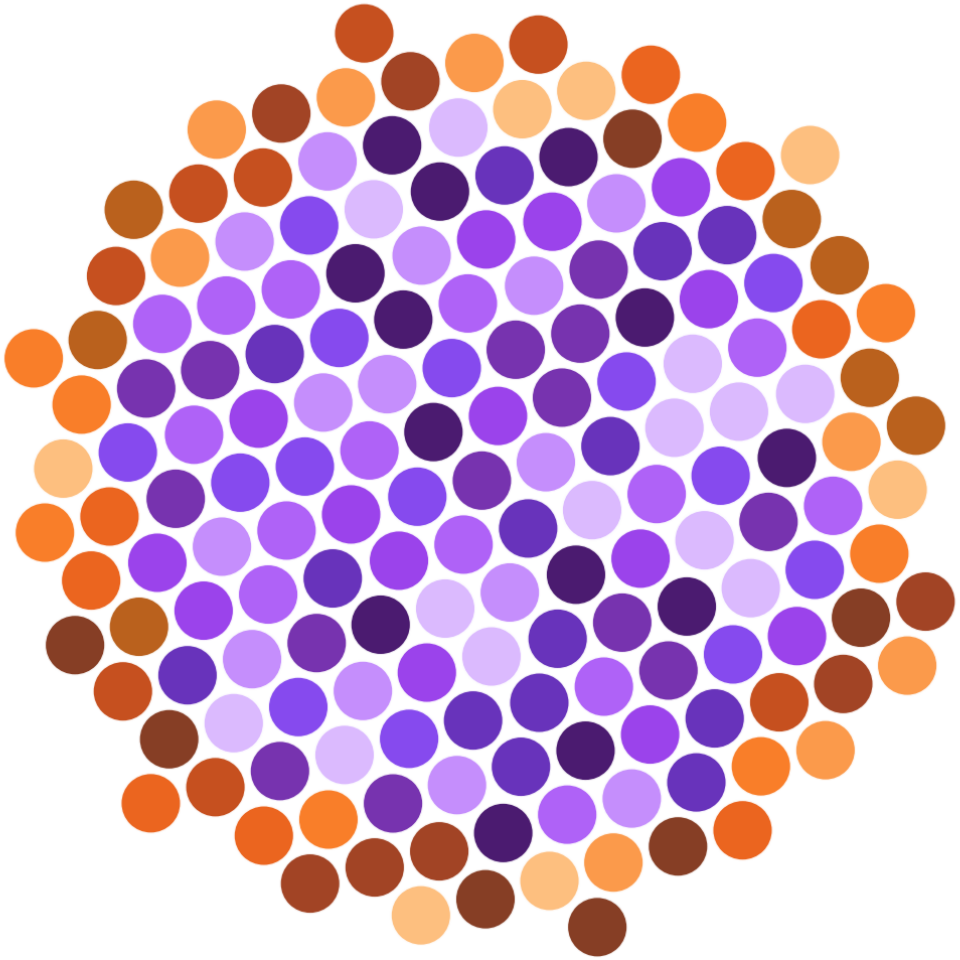
2. **Do not ignore the elephant in the room! Go *beyond coding* and teach about the gender gap, stereotypes, counterstereotypical examples, …**

Image source: Wikimedia Commons

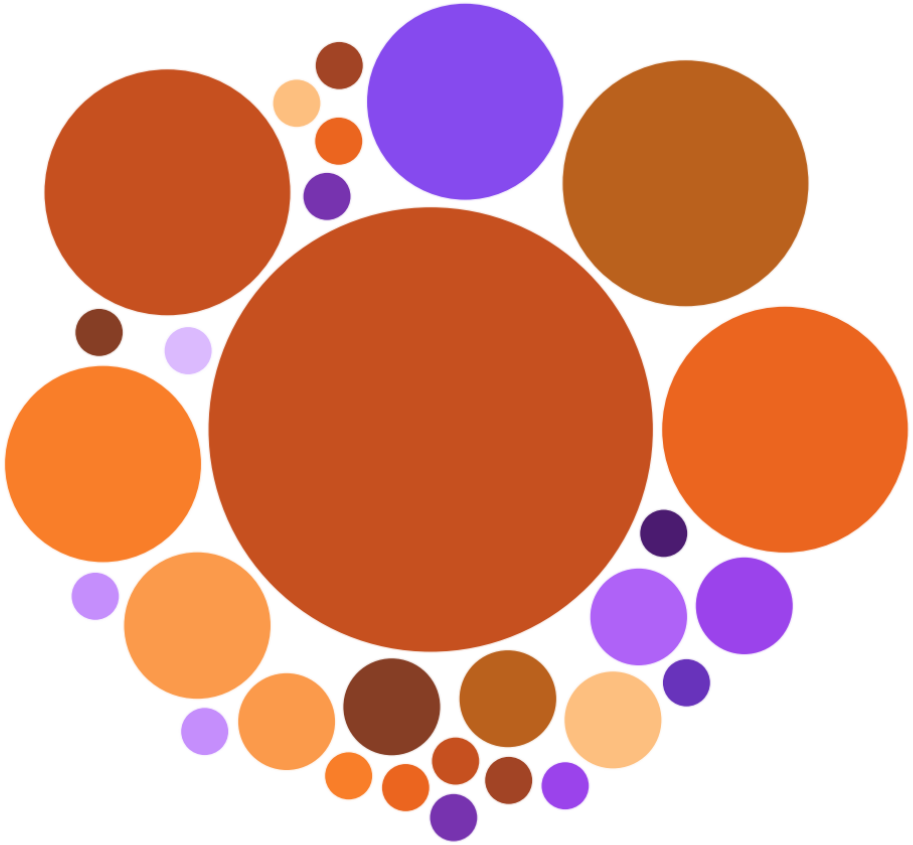
1. **Early exposure to programming.**
2. **Do not ignore the elephant in the room! Go *beyond coding* and teach about the gender gap, stereotypes, counterstereotypical examples, …**
3. **Diversify role models.**

Who studies a Ph.D. in psychology?



Women (110) Men (60)

Who teaches programming to psychologists?



Women (11) Men (18)

Data from the **University of Granada** Ph.D. Program in Psychology and the Mind, Brain and Behavior Research Center.

1. **Early exposure to programming.**
2. **Do not ignore the elephant in the room! Go *beyond coding* and teach about the gender gap, stereotypes, counterstereotypical examples, …**
3. **Diversify role models.**
4. **Fight barriers widening the *real* access.**

Programming in Psychological Science A Practical Introduction

Start reading! View on GitHub

https://wobc.github.io/programming_book/

This guide is **not** a full programming course or a replacement for formal tutorials. Instead, it's a **starting point**: a collection of reflections, examples, and recommendations from the perspective of researchers in psychology and cognitive neuroscience. It aims to help you get your bearings and make smart decisions as you begin learning to code — not to teach you everything from scratch.

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Maria Jesús Maraver

Javier Ortiz-Tudela

Ana F. Palenciano

Rodika Sokoliuk

[SheCodesPsy]: Reprogramming stereotypes!

3-day immersive course designed for female+ psychology students to learn basic Python programming. No inscription fee, free accommodation for international students.

Where and when? Mind, Brain and Behavior Research Center (CIMCYC), University of Granada (Spain). 13-15 May 2026.

https://shecodespsy.github.io/

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Thanks for listening - *and for caring!* 😌

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